Mathematical Foundations of the AI Web Economy Simulator

A Lotka-Volterra Model with Mechanism Design Constraints

Version 3.0

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Abstract

This note presents the mathematical foundations of an ecosystem model for AI platforms and web content creators. The model uses a "one-mechanism-per-layer" structure: each causal channel appears exactly once, making the system auditable and preventing double-counting. Creator dynamics respond to profitability, traffic responds to AI substitution pressure, quality responds to creator mass and reinvestment, and AI growth depends on content quality. All state variables are bounded in $[0, 1]$ by construction.

1 Model Structure

1.1 Design Principle: One Mechanism Per Layer

The model follows a strict causal chain where each variable is responsible for one layer of the ecosystem:

$$\begin{array}{ccccccccc} A \text{ grows} & \rightarrow & W \text{ drops} & \rightarrow & \pi_c \text{ drops} & \rightarrow & C \text{ drops} & \rightarrow & Q \text{ drops} \\ & & & & & & & & \downarrow \\ & & & & & & & & \text{feeds back to } A \end{array}$$

- A does not directly hurt creators—it hurts them *through* W (traffic diversion)
- W does not directly change C—it changes C *through* π_c (revenue)
- π_c does not directly change C—it changes C *through* the entry/exit decision
- C does not directly change Q—it changes Q *through* critical mass and reinvestment
- Q feeds back to A *through* the data advantage channel (better content $\rightarrow$ better AI)

No mechanism appears twice. Every effect can be traced through a single logical path.

1.2 State Variables

Definition 1 (State Variables)**.** *The model tracks four state variables, all normalized to* $[0, 1]$*:*

$$C(t) : \textit{Creator participation (active creator mass, normalized)} \tag{1}$$
$$A(t) : \textit{AI platform capability/scale (normalized)} \tag{2}$$
$$W(t) : \textit{Traffic share to original content (vs. staying in AI answers)} \tag{3}$$
$$Q(t) : \textit{Original content quality index} \tag{4}$$

Remark 1. *$W = 1$ means all searches click through to original sources; $W = 0$ means AI answers all queries with no click-through. Q refers exclusively to* original *content quality, not AI output quality.*

2 Core Dynamics

Assumption 1 (Nonnegative Rates)**.** *All rate parameters are nonnegative:* $\rho_C, \rho_W, \nu, \epsilon, \lambda_0, I, r_A, \delta_A \geq 0$*. This ensures the bounded gain–loss forms keep state variables in* $[0, 1]$ *by construction.*

All equations use bounded "gain–loss" forms that keep state variables in $[0, 1]$ by construction, without requiring ex post clamping.

2.1 Creator Revenue

$$\boxed{\pi_c = \theta \cdot p_L \cdot A + p_{ad} \cdot W + p_{sub} \cdot Q} \tag{5}$$

Interpretation:

- $\theta \cdot p_L \cdot A$: Licensing revenue from AI platforms (scales with compensation rate θ and AI scale A)
- $p_{ad} \cdot W$: Advertising revenue (proportional to traffic reaching original content)
- $p_{sub} \cdot Q$: Subscription/patronage revenue (scales with content quality)

Remark 2. *Traffic diversion affects revenue* only *through* W*. There is no additional* $(1 - \alpha A)$ *multiplier—that would double-count diversion.*

2.2 Creator Dynamics (Entry/Exit)

Creators enter or exit based on whether profitability exceeds their outside option $\bar{u}$. We use a sigmoid response function:

$$S(x) = \frac{1}{1 + e^{-\beta x}} \tag{6}$$

where $\beta > 0$ controls response steepness. (We later use $\sigma(x) = 1/(1 + e^{-x})$ for diversion dynamics; the distinction is that S has adjustable slope β for calibrating creator sensitivity, while σ has unit slope with sensitivity absorbed into k_A, k_θ.)

The bounded creator dynamics are:

$$\boxed{\frac{dC}{dt} = (1 - C) \cdot \rho_C \cdot S(\pi_c - \bar{u}) - C \cdot \rho_C \cdot S(\bar{u} - \pi_c)} \tag{7}$$

Interpretation:

- $(1 - C) \cdot \rho_C \cdot S(\pi_c - \bar{u})$: Entry rate—new creators join when profitable
- $C \cdot \rho_C \cdot S(\bar{u} - \pi_c)$: Exit rate—existing creators leave when unprofitable
- The $(1 - C)$ and C terms ensure $C \in [0, 1]$ by construction
- The IR constraint emerges from the dynamics: when $\pi_c < \bar{u}$, exit dominates entry

Remark 3. *This replaces indicator-function regime penalties (κ_1, κ_2). The constraint is now* operational *rather than bolted on.*

2.3 Traffic Dynamics

Traffic to original content depends on AI substitution pressure, moderated by compensation:

Definition 2 (Diversion Propensity)**.** *Define $D(A, \theta) \in [0, 1]$ as the propensity for users to stay inside AI answers:*

$$D(A, \theta) = \sigma\Big(k_A(A - a_0) - k_\theta(\theta - \theta_0)\Big) \tag{8}$$

where $\sigma(x) = 1/(1 + e^{-x})$ is the logistic function, with $k_A, k_\theta > 0$.

Interpretation:

- D increases with AI scale A (more capable AI $\rightarrow$ more substitution)
- D decreases with compensation θ (higher compensation $\rightarrow$ platforms link to sources)

The bounded traffic dynamics are:

$$\boxed{\frac{dW}{dt} = (1 - W) \cdot \rho_W \cdot (1 - D) - W \cdot \rho_W \cdot D} \tag{9}$$

Interpretation:

- When D is high (strong substitution), W decreases toward 0
- When D is low (weak substitution), W increases toward 1
- The $(1 - W)$ and W terms ensure $W \in [0, 1]$ by construction

Remark 4. *Q does not appear in dW/dt. This is a design choice (Option A): traffic diversion depends only on AI capability and compensation, not on content quality. This keeps the causal chain maximally identifiable.*

2.4 Content Quality Dynamics

Quality depends on creator mass (critical mass effect) and reinvestment from profitable creators:

Definition 3 (Reinvestment Signal)**.** *Define the reinvestment signal as a smooth function of excess profitability:*

$$R(\pi_c) = \ln(1 + e^{\pi_c - \bar{\pi}}) \tag{10}$$

This is the softplus function, a smooth approximation to $\max(0, \pi_c - \bar{\pi})$.

The bounded quality dynamics are:

$$\boxed{\frac{dQ}{dt} = (1 - Q)\Big[\nu \cdot R(\pi_c) + \epsilon \cdot \max(C - C_{crit}, 0)\Big] - Q\Big[\lambda_0 + \epsilon \cdot \max(C_{crit} - C, 0)\Big]} \tag{11}$$

Interpretation:

- $(1 - Q)\nu R(\pi_c)$: Reinvestment-driven quality improvements, bounded as $Q \rightarrow 1$
- $(1 - Q)\epsilon \max(C - C_{crit}, 0)$: Critical mass upside only applies when quality is not already maxed
- $-Q[\lambda_0 + \epsilon \max(C_{crit} - C, 0)]$: Decay and critical mass shortfall only reduce quality in proportion to existing quality (prevents $Q < 0$)

Remark 5. *There is no direct θ term in dQ/dt. The effect of low compensation on quality flows through the chain: low $\theta \rightarrow$ low $\pi_c \rightarrow$ low $R \rightarrow$ quality doesn't improve $\rightarrow$ and low $\pi_c \rightarrow$ creator exit $\rightarrow C < C_{crit} \rightarrow$ quality declines.*

Remark 6. *The* $\max(\cdot, 0)$ *terms introduce non-differentiability at $C = C_{crit}$, making the system piecewise-smooth. For simulation this is unproblematic; for rigorous stability analysis, these can be replaced with softplus approximations.*

2.5 AI Platform Dynamics

AI platforms grow with investment and content availability. Quality enters here via the **data advantage channel**: better original content improves AI training and grounding.

$$\boxed{\frac{dA}{dt} = (1 - A) \cdot \Big(I + r_A \cdot A \cdot (1 + \gamma Q)\Big) - A \cdot \delta_A} \tag{12}$$

Interpretation:

- I: External investment (exogenous inflow)
- $r_A \cdot A \cdot (1 + \gamma Q)$: Growth from existing capability, *amplified* by content quality Q
- $(1 - A)$: Saturation as $A \to 1$
- $A \cdot \delta_A$: Depreciation/competition

Remark 7. *The* $(1+\gamma Q)$ *term is multiplicative on* endogenous *growth, not additive. This means low* Q *throttles the endogenous growth component* $r_A A(1 + \gamma Q)$*, though exogenous investment* I *can still drive* A *forward independently—encoding the claim that "AI benefits from quality content" while allowing for investment-driven growth.*

3 Mechanism Design Constraints

The IR and IC constraints are now *emergent properties* of the dynamics rather than separate conditions.

3.1 Individual Rationality (IR)

Definition 4 (Participation Constraint)**.** *Creators participate when expected revenue exceeds their outside option:*

$$IR: \quad \pi_c \geq \bar{u} \tag{13}$$

This constraint is enforced by the entry/exit dynamics in Section 2.2. When $\pi_c < \bar{u}$, exit rate exceeds entry rate, and C declines.

3.2 Incentive Compatibility (IC)

Definition 5 (Quality Incentive Constraint)**.** *High-quality content is incentive compatible when the subscription/patronage premium exceeds additional costs:*

$$IC: \quad p_{sub} \cdot \Delta Q \geq \Delta c \tag{14}$$

where ΔQ *is the quality increment from low to high, and* $\Delta c = c_{high} - c_{low}$ *is the* ***incremental cost of quality****—the extra investment required for research, fact-checking, editing, and original reporting versus quick, low-effort content production.*

Under the $[0, 1]$ normalization, we take $\Delta Q = 1$ for the low-to-high quality shift. This means IC simplifies to $p_{sub} \geq \Delta c$: creators invest in quality only when the subscription premium exceeds the additional production cost.

Remark 8. *IC is a* ***reduced-form interpretation****, not an explicit regime switch in the ODEs. The dynamics do not contain a discrete "quality choice" variable. Instead, IC violation manifests through the revenue channel: when* p_{sub} *is low,* Q *contributes less to* π_c*, which reduces* $R(\pi_c)$*, which slows quality improvement in Eq. (11). The constraint thus operates implicitly through the continuous dynamics rather than as an explicit threshold.*

4 Parameters

Symbol	Description	Role	Units
Revenue parameters			
θ	Compensation rate	Policy lever	$[0, 1]$
p_L	Licensing price coefficient	AI willingness to pay	\$/unit
p_{ad}	Ad revenue coefficient	Revenue per traffic unit	\$/unit
p_{sub}	Subscription coefficient	Revenue per quality unit	\$/unit
$\bar{u}$	Outside option (reservation utility)	IR threshold	\$
$\bar{\pi}$	Reinvestment threshold	IC proxy	\$
Creator dynamics			
ρ_C	Creator adjustment speed	Entry/exit rate	1/time
β	Profitability response steepness	Sigmoid sharpness	1/\$
Traffic dynamics			
ρ_W	Traffic adjustment speed	Diversion rate	1/time
k_A	AI scale sensitivity	Diversion response to A	dimensionless
k_θ	Compensation sensitivity	Diversion response to θ	dimensionless
a_0, θ_0	Baseline thresholds	Centering parameters	dimensionless
Quality dynamics			
ν	Quality improvement rate	Reinvestment response	1/time
ϵ	Critical mass sensitivity	Quality response to C	1/time
C_{crit}	Critical creator mass	Threshold parameter	dimensionless
λ_0	Baseline quality decay	Depreciation rate	1/time
AI dynamics			
I	External investment rate	Exogenous inflow	1/time
r_A	AI growth rate	Endogenous scaling	1/time
γ	Data advantage strength	AI quality dependence	dimensionless
δ_A	AI depreciation rate	Competition/obsolescence	1/time
IC constraint parameters			
ΔQ	Quality increment (low to high)	Normalization	dimensionless (=1)
Δc	Incremental cost of quality	IC threshold	\$

5 Equilibrium Analysis

5.1 Steady State Conditions

At equilibrium, all time derivatives are zero. The system admits multiple equilibria depending on parameters.

Traffic equilibrium: Setting $dW/dt = 0$:

$$W^* = 1 - D(A^*, \theta) \tag{15}$$

Traffic share equals one minus diversion propensity.

Creator equilibrium: Setting $dC/dt = 0$:

$$(1 - C^*)S(\pi_c^* - \bar{u}) = C^* S(\bar{u} - \pi_c^*) \tag{16}$$

Using $S(\bar{u} - \pi_c) = 1 - S(\pi_c - \bar{u})$ for the logistic, the interior steady-state condition is:

$$C^* = S(\pi_c^* - \bar{u}) \tag{17}$$

Equivalently: $\pi_c^* = \bar{u} + \frac{1}{\beta}\ln\left(\frac{C^*}{1-C^*}\right)$.

At interior equilibrium, creator participation equals the sigmoid of profitability relative to the outside option.

AI equilibrium: Setting $dA/dt = 0$:

$$(1 - A^*)\big(I + r_A A^*(1 + \gamma Q^*)\big) = A^* \delta_A \tag{18}$$

This is implicit in A^*. A first-order approximation when A is small:

$$A^* \approx \frac{I}{\delta_A} \quad \text{(first-order, small-}A\text{ approximation)} \tag{19}$$

In the small-A regime, $A^* \approx I/\delta_A$ and Q affects A^* only weakly; for larger A, Q materially shifts A^* through the endogenous growth term.

5.2 Equilibrium Types

Proposition 1 (Possible Equilibria)**.** *The system admits three qualitatively distinct equilibria:*

1. ***Sustainable Coexistence:*** *$C^* > C_{crit}$, $Q^* > 0$, $W^* > 0$. Both IR and IC effectively satisfied; stable ecosystem.*
2. ***Low-Quality Trap:*** *$C^* \approx C_{crit}$, Q^* settles at a low level. Creators survive but quality erodes.*
3. ***Collapse:*** *$C^* \to 0$, $Q^* \to 0$. Creator exodus, quality collapse, AI faces data drought.*

5.3 Stability

For the 4-dimensional system (C, A, W, Q), the bounded gain–loss forms prevent state blow-ups and keep trajectories within $[0,1]^4$. Local stability at each equilibrium is determined by the eigenvalues of the Jacobian evaluated at that point.

6 Policy Implications

6.1 Compensation as the Key Lever

The compensation rate θ affects the system through exactly two channels:

1. **Revenue channel:** Higher θ increases π_c directly via $\theta \cdot p_L \cdot A$
2. **Diversion channel:** Higher θ reduces $D(A, \theta)$, preserving traffic W

Both channels reinforce each other: higher compensation both increases direct licensing revenue and preserves the ad-supported traffic channel.

6.2 Critical Thresholds

Two thresholds emerge from the dynamics:

1. **Survival threshold θ^{IR}:** The minimum compensation at which $\pi_c = \bar{u}$ can be achieved. Below this, creators cannot survive.
2. **Quality threshold θ^{IC}:** The minimum compensation at which reinvestment is viable ($R(\pi_c) > 0$ meaningfully). Below this, quality erodes.

These thresholds depend on the full parameter vector and must be computed numerically for specific calibrations.

7 Summary

The model implements a clean causal chain:

$$A \uparrow \to\ W \downarrow \to\ \pi_c \downarrow \to\ C \downarrow \to\ Q \downarrow \to\ A \text{ (slows)}$$

Each mechanism appears exactly once:

- Diversion lives in W (and only W)
- Revenue depends on W (and only W) for the traffic channel
- Creator dynamics respond to profitability (and only profitability)
- Quality responds to creator mass and reinvestment (and only those)
- AI growth depends on quality (and only quality) for the feedback

This structure ensures the model is auditable, calibratable, and produces interpretable policy counterfactuals.

Version 3.0. Feedback welcome.